

SHORT-CIRCUIT CALCULATIONS

Per Unit Impedance Method:

The per-unit (p.u.) impedance method simplifies short-circuit calculations in power systems by normalizing component impedances and quantities to a common base value, this method involves converting actual system impedances to a dimensionless p.u. value by dividing by a chosen base impedance, which is derived from selected base voltage (kV) and apparent power (MVA) values.

By representing all system impedances on a common scale, engineers can construct a per-unit impedance diagram to find the total system impedance up to a fault location and then calculate the short-circuit current and MVA.

Per unit Impedance method:

Formulas used:

- $Z \text{ per unit} = Z \text{ Actual} / Z \text{ Base}$
- $Z \text{ per unit} = Z \text{ Actual} * \text{MVA (New)} / \text{MVA (Old)}$
- For Cables $Z \text{ per unit} = \text{MVA (New)} / \text{kV}^2$
- $\text{Fault MVA} = \text{Base MVA} / Z \text{ per unit}$

Let go with a theoretical example to see how the method works:

Given data:

T2 = 50 MVA, 132/33kV, %Z = 10%

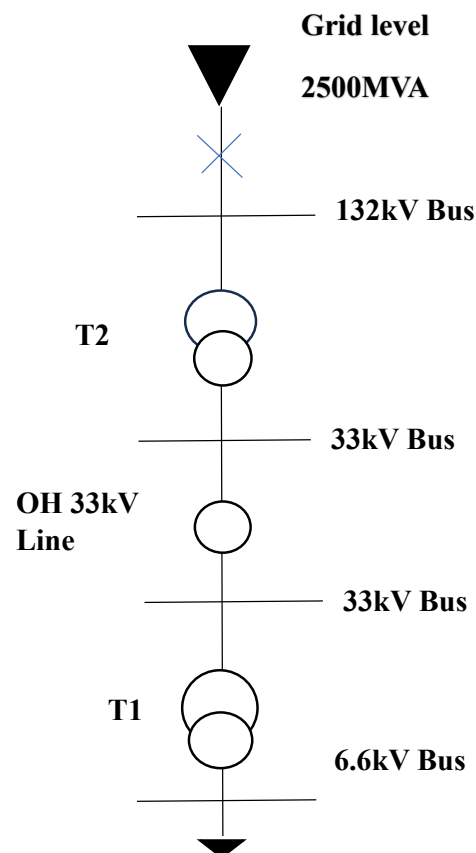
OH 33kV 3 KM length, $0.4 \Omega / \text{KM}$

T1 = 8 MVA, 33/6.6kV, %Z = 8%

Now, consider Base MVA = 100 MVA

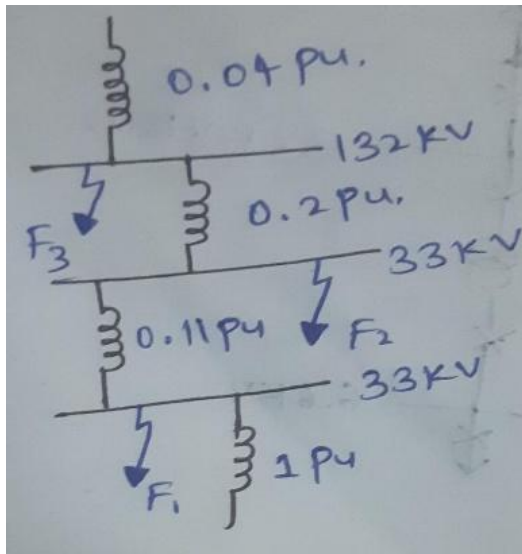
consider, $Z \text{ system} = 1.0$

- $Z \text{ per unit of Grid system,}$
 $Z \text{ act} * \text{MVA (New)} / \text{MVA (old)}$
 $= 1.0 * 100 / 2500$
 $= 0.04 \text{ P. U}$
- $Z \text{ per unit of T1,}$
 $Z \text{ act} * \text{MVA (New)} / \text{MVA (old)}$
 $= 0.1 * 100 / 50$
 $= 0.2 \text{ P. U}$



- Z per unit of OH line,
 $Z_{act} = 0.4 * 3 = 1.2 \text{ } \Omega$
 $Z (P. U) = Z_{act} * MVA (New) / kV^2$
 $= 1.2 * 100 / (33^2)$
 $= 0.11 P. U$
- Z per unit of T2,
 $Z_{act} * MVA (New) / MVA (Old)$
 $= 0.08 * 100 / 8$
 $= 1 P. U$

Impedance drawing:



- Calculation of Fault F1 @33kV bus,

If there is a fault f1 the whole system is parallel contributors with T1,

$$Z_{equivalent} = [0.04 + 0.2 + 0.11] // [1]$$

$$= 0.259 p. u$$

$$\text{Fault MVA} = \text{base MVA} / Z_{p.u}$$

$$= 100 / 0.259$$

$$= 386.100 \text{ MVA}$$

$$\text{Short-circuit current at Fault 1} = I(f1) = S / \sqrt{3} * V$$

$$= 386.1 \text{ MVA} / \sqrt{3} * 33 \text{ kV}$$

$$I(f1) = 6.75 \text{ kA}$$

- Calculation of fault 2 @33kV OH line,

If the fault f2 at OH line the whole system upside the OH line is act as parallel contributors, T1 will act as series contributor,

$$Z_{\text{equivalent}} = [0.04+0.2] // [0.11+1]$$

$$= 0.197 \text{ p. u}$$

$$\text{Fault MVA} = \text{base MVA} / Z \text{ p. u}$$

$$= 100 / 0.197$$

$$= 507.614 \text{ MVA}$$

$$\text{Short-circuit current at Fault 2} = I(f2) = S / \sqrt{3} * V$$

$$= 507.614 \text{ MVA} / \sqrt{3} * 33\text{kV}$$

$$I(f2) = 8.88 \text{ kA}$$

- Calculation of fault 3 @132 kV bus,

If the fault f3 at T2 at 132kV bus the remaining whole system act as parallel contributors to T2,

$$Z_{\text{equivalent}} = [0.04] // [0.2+0.11+1]$$

$$= 0.038 \text{ p. u}$$

$$\text{Fault MVA} = \text{base MVA} / Z \text{ p. u}$$

$$= 100 / 0.038$$

$$= 2631.578 \text{ MVA}$$

$$\text{Short-circuit current at Fault 3} = I(f3) = S / \sqrt{3} * V$$

$$= 2631.578 \text{ MVA} / \sqrt{3} * 132\text{kV}$$

$$I(f3) = 11.510 \text{ kA}$$

Hence, I have calculated fault MVA, and fault current ratings at each feeder line by using **PER UNIT IMPEDANCE METHOD** of short-circuit calculations.